

The CTC Algorithm — Solutions Manual

manim-concepts

Chapter 1

Counting rules

Practice problems

Problem 1.1. [Calibration placeholder — replaced in phase 2 under the solve gate.] Three lights each show one of three colours, independently. How many distinct displays can the panel show?

Solution. Each light contributes a factor of 3 and the stages are independent, so the multiplicative rule gives $3 \times 3 \times 3 = 27$ distinct displays. (Same shape as the raw CTC path count $3^4 = 81$ the alignment chapter builds on this rule.)